C2 Research Logbook

Audio to synthesizer parameter estimation and reliability calibration

Student name _____ Student ID IT23203280

Record period 03 August to 20 October 2026 Compiled 20 October 2026

Entries group related work by date. Weekly review comments and signatures are completed by the reviewer using the actual signing date.

Week 1 03 to 09 August 2026

Initial setup and audio generation

Date	Work completed and result
03 Aug	Set up the Python environment, repository and Vital/Vita rendering tools. Investigated the initial Vita compatibility issue. **Result** Established the first working route for generating synthesizer audio.
04 Aug	Generated and validated the initial 1,024-sample pilot. Checked mono output, file count, sample rate and duration. **Result** Saved an initial dataset and validation outputs for pipeline development.
06-07 Aug	Reviewed the first audio examples and the generation settings. Listed sound-quality issues to investigate. **Result** Made audio quality and consistent rendering the immediate priorities.

Weekly outcome

The initial generator and pilot were available for further technical checks.

Next step

Check audio consistency and the relationship between recorded settings and sound.

Weekly review

Reviewer comments

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Reviewed by _____ Role Supervisor / Co supervisor

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Week 2 10 to 16 August 2026

Basic audio checks and parameter records

Date	Work completed and result
10-11 Aug	Reviewed WAV handling, mono conversion and note timing. Checked that each clip had a corresponding parameter record. **Result** Clarified the inputs and labels needed for later model training.
13 Aug	Reviewed possible clipping, quiet output and incomplete release tails in generated audio. **Result** Separated audio-generation concerns from later model-performance concerns.
16 Aug	Reviewed the first two weeks of setup and pilot work. Collected the main issues and settings in project notes. **Result** Kept a short list of diagnostics needed before expanding the pipeline.

Weekly outcome

The audio checks and parameter-record requirements were clearer.

Next step

Make the rendering and validation process easier to repeat.

Weekly review

Reviewer comments

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Week 3 17 to 23 August 2026

Rendering consistency and dataset organisation

Date	Work completed and result
17-18 Aug	Reviewed the eight pilot controls and their allowed ranges. Kept the note, velocity and recording duration fixed. **Result** Reduced avoidable variation between examples.
20 Aug	Reviewed file naming, sample identities and the manifest used to link audio with synthesizer settings. **Result** Established a consistent basis for selecting and loading examples.
21-23 Aug	Checked the first-version workflow and documented the remaining generation and storage issues. **Result** Prepared the pilot workflow for repeatable preprocessing and training.

Weekly outcome

The first version had a clearer generation recipe and dataset structure.

Next step

Finish storage and progress tracking, then run the initial estimation baseline.

Weekly review

Reviewer comments

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Week 4 24 to 30 August 2026

Pilot stabilisation and first estimation baseline

Date	Work completed and result
24 Aug	Updated pilot storage to PCM16 and set the generation checkpoint interval to 1,024 samples. **Result** Made the small pilot easier to store and resume.
25 Aug	Added generation progress reporting and checked the canonical pilot configuration. **Result** Made generation progress visible and kept the dataset settings recorded.
26 Aug	Ran the first C2 baseline using log-mel inputs and a small CNN for eight controls. Used 819 training, 102 validation and 103 test clips. **Result** Saved the model, spectrograms, learning curves and per-control MAE, RMSE and R squared.
28-30 Aug	Reviewed the pilot results and reproducibility checks. Attack and release were recovered better than resonance and drive. **Result** Recorded the weaknesses of the small-data baseline rather than relying only on its average score.

Weekly outcome

The first version connected validated pilot audio to a repeatable estimation baseline.

Next step

Consolidate the first version and define the research work needed for C2.

Weekly review

Reviewer comments

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Week 5 31 August to 06 September 2026

First-version review and transition to C2

Date	Work completed and result
31 Aug-01 Sep	Consolidated notes on the pilot, audio checks and baseline outputs. **Result** Kept the first version as a reference for later changes.
02-03 Sep	Reviewed preprocessing, fixed splits and saved metrics from the initial pipeline. **Result** Completed the first-version review and identified its research limitations.
05 Sep	Shifted the main focus to C2. Defined the need to estimate synthesizer controls and explain when those estimates are reliable. **Result** Set parameter estimation and reliability calibration as the component's main tasks.
06 Sep	Outlined the C2 workflow from audio features to predicted controls, reconstruction and reliability assessment. **Result** Retained the small CNN as the starting baseline.

Weekly outcome

C2 became the main focus after the first-version setup and diagnostic work.

Next step

Develop the uncertainty and acoustic-identifiability rationale.

Weekly review

Reviewer comments

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Week 6 07 to 13 September 2026

C2 scope and uncertainty design

Date	Work completed and result
07-08 Sep	Examined why different synthesizer settings can produce similar sounds and why some controls may be difficult to recover. **Result** Defined parameter-recovery reliability separately from perceived sound similarity.
10-11 Sep	Reviewed the pilot model's stronger and weaker controls and the limits of reporting one overall error. **Result** Made per-control evaluation a requirement for the extended component.
12-13 Sep	Outlined the use of several independently trained models to estimate uncertainty. **Result** Planned an ensemble comparison while keeping the deterministic baseline for reference.

Weekly outcome

The component's uncertainty question was linked to observable weaknesses in parameter recovery.

Next step

Define reliability metrics and the forward-rendering checks.

Weekly review

Reviewer comments

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Week 7 14 to 20 September 2026

Evaluation planning and progress review

Date	Work completed and result
14-15 Sep	Consolidated the completed first-version work and the proposed C2 direction in the project progress report. **Result** Separated implemented results from the planned uncertainty work.
17-18 Sep	Reviewed probability-quality and calibration measures alongside MAE, RMSE and R squared. **Result** Identified Brier score, calibration plots and risk-coverage analysis for the reliability stage.
20 Sep	Reviewed the previous two weeks of C2 planning and the evidence needed to justify its claims. **Result** Kept acoustic ambiguity and the need for independent validation explicit.

Weekly outcome

The evaluation plan covered both point predictions and the quality of reliability estimates.

Next step

Specify the uncertainty, sensitivity and reconstruction comparisons.

Weekly review

Reviewer comments

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Week 8 21 to 27 September 2026

Forward diagnostics and reliability comparisons

Date	Work completed and result
21-23 Sep	Outlined local parameter perturbations around a predicted patch and comparison of the resulting audio. **Result** Planned sensitivity measurements without changing the delivered prediction.
24-25 Sep	Planned comparisons using uncertainty alone, uncertainty with sensitivity, uncertainty with reconstruction error, and all three together. **Result** Made the contribution of each reliability input testable.
27 Sep	Reviewed the proposed perturbation workload and the distinction between training data and diagnostic examples. **Result** Chose to investigate fixed subsets rather than perturb the entire corpus.

Weekly outcome

The C2 design had a concrete comparison strategy for its reliability inputs.

Next step

Plan protected data partitions and the parameter-correctness event.

Weekly review

Reviewer comments

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Week 9 28 September to 04 October 2026

Validation protocol and research design review

Date	Work completed and result
28-30 Sep	Planned separate roles for training, calibration, model selection and final testing. **Result** Established the need to prevent reliability fitting and final evaluation from sharing examples.
01-02 Oct	Outlined reliability as the chance of recovering a control within a declared tolerance in its research range. **Result** Defined a measurable event for calibration rather than treating raw variance as a probability.
04 Oct	Reviewed two weeks of diagnostic and validation planning. Identified the need to revisit the synthesis domain before scaling the work. **Result** Set parameter-domain checks and a larger controlled corpus as the next dependencies.

Weekly outcome

The proposed protocol linked each metric and data subset to a specific research purpose.

Next step

Confirm the control set and audio-processing policy.

Weekly review

Reviewer comments

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Week 10 05 to 11 October 2026

Research domain and audio conditioning

Date	Work completed and result
05-06 Oct	Reviewed the expanded single-oscillator domain and the roles of the amplitude and filter envelopes. **Result** Moved beyond the initial eight-control engineering pilot.
07 Oct	Selected the final 13-control domain, retaining Bend and filter-envelope sustain. Investigated DC offset and compared mitigation options. **Result** Selected a fixed 10 Hz high-pass rule, preserving raw audio and recording the processing policy.
08-09 Oct	Reviewed storage precision and quiet spectral energy, together with generation and resume requirements. **Result** Kept FLOAT research masters and separate PCM16 playback copies.
10-11 Oct	Prepared the larger-corpus workflow, fixed parameter ranges and reproducibility requirements for C2. **Result** Defined the data and runtime checks needed before full model training.

Weekly outcome

C2 had a specified 13-control domain and a consistent audio-conditioning rule.

Next step

Complete the corpus checks, freeze C2 partitions and train the full models.

Weekly review

Reviewer comments

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Week 11 12 to 18 October 2026

Full C2 training and development validation

Date	Work completed and result
12-13 Oct	Completed checks for the 131,072-clip v4 corpus, including the FLOAT resume issue. Froze 98,304 training, 16,384 calibration, 8,192 validation and 8,192 test assignments. **Result** Split validation equally between inverse-model and reliability selection; kept the earlier 1,024 v4 clips in training.
13 Oct	Trained the deterministic 13-output CNN and saved data distributions, spectrograms, learning curves and per-control metrics. **Result** Measured inverse-validation macro z MAE of 0.126112 and retained the saved model and predictions.
14 Oct	Completed the three-member ensemble and diagnostics on 2,048 calibration and 512 reliability-validation clips. **Result** Ensemble MAE was 0.127991. Brier was 0.116267 for uncertainty alone and 0.115385 for all three inputs.
14-15 Oct	Audited the reliability comparisons and tested perturbation steps 0.01, 0.02 and 0.05 on 256 fixed calibration examples. **Result** Retained step 0.02. The audit did not establish an extra sensitivity benefit beyond uncertainty plus reconstruction error.
15 Oct	Built the saved-results viewer and organised the technical and demonstration guides. **Result** Made target/reconstruction audio, controls, reliability estimates and saved evidence easier to inspect.
18 Oct	Built and checked the separate C2 human-study rehearsal with one practice case, eight scored cases and counterbalanced reliability displays. **Result** Prepared a target of 12 completions with a 16-enrolment cap. Five automated tests passed; research collection remained disabled.

Weekly outcome

The development pipeline, reliability audit and study rehearsal were available. The 13-control final test remained reserved.

Next step

Confirm the human-study materials and approvals, then lock the method before final testing.

Weekly review

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Week 12 19 to 20 October 2026

Documentation and remaining validation work

Date	Work completed and result
20 Oct	Organised the C2 research record and prepared this logbook. Reviewed the saved outputs, study design and remaining validation tasks. **Result** Kept development results, study preparation and pending final evaluation clearly distinguished.

Weekly outcome

The C2 work was documented for review through 20 October 2026.

Next step

Rehearse and time the study, finalise approved materials, and complete the remaining evaluation stages.

Weekly review

Reviewer comments

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